

NIVARA

Backend Product Requirements Document (PRD)

Version	Project	Primary Roles
2.0	Early Student Support Ecosystem	Student, Counsellor, Administrator

1. Product Vision

Nivara is a student-support platform that identifies students who **may need academic, financial, or well-being support** using minimal and consented data.

Instead of waiting until a student reaches a crisis point, Nivara detects meaningful changes in student circumstances and connects them with appropriate support.

Support before the breaking point.

The system does **not** diagnose, punish, rank, or automatically make high-impact decisions about students.

Its purpose is to:

Identify → Explain → Support → Connect → Follow Up → Improve

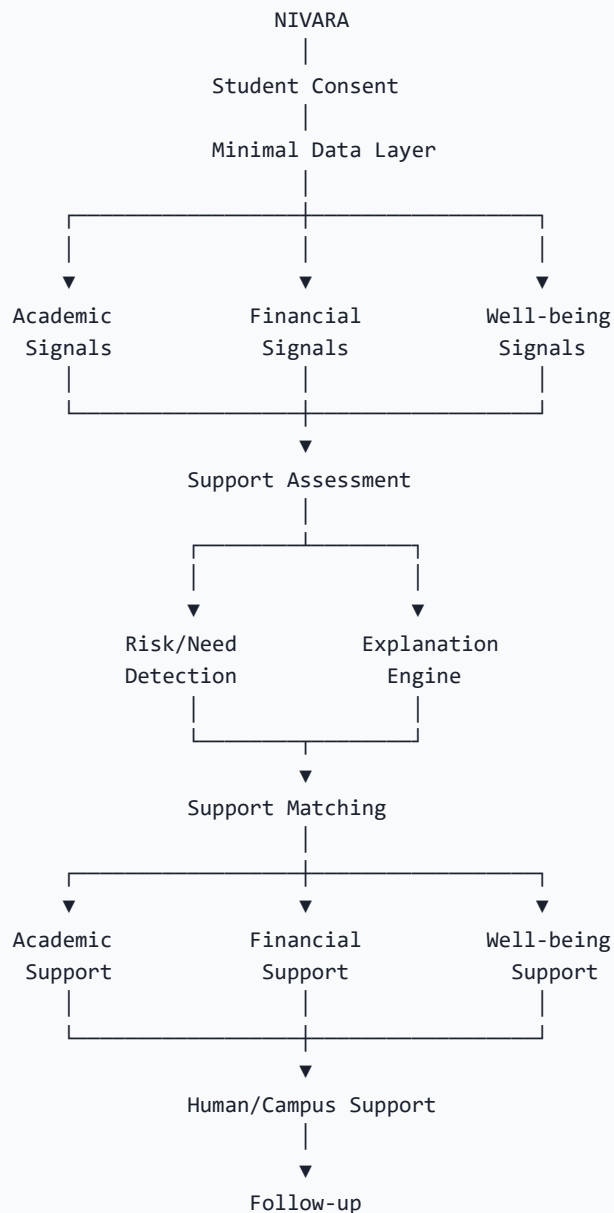
2. Problem Statement Interpretation

PS-29 requires the system to:

1. **Identify** students who may need: Academic support, Financial support, Well-being support
2. **Use**: Minimal data, Consent-based data collection
3. **Explain**: Why a student may need support, Which factors contributed to the recommendation
4. **Provide**: Appropriate intervention/support options
5. **Measure**: Bias across student groups
6. **Prevent**: Automated punitive decisions
7. **Allow students to**: View key profile information, Correct inaccurate information

Nivara's backend is designed around these seven requirements.

3. Core Nivara Architecture



4. Three Core Support Pillars

Nivara will organize student support into three primary categories.

4.1 Academic Support

Signals may include:

- Self-reported academic difficulty
- Academic stress
- Missed assignments
- Declining attendance, where consented and institutionally permitted

- Difficulty understanding subjects
- Examination pressure
- Placement-related anxiety

Potential interventions: Academic advisor, Faculty support, Peer tutoring, Study resources, Mentoring, Time-management resources, Exam preparation resources.

4.2 Financial Support

Financial support is a **support-navigation system**, not a financial judgment system. Nivara should never label a student as "poor" or financially unfit.

"You indicated that you may be experiencing difficulty with educational/living expenses. Here are support options you may want to explore."

Potential support options: Scholarships, Fee assistance, Fee installment plans, Emergency student funds, Hostel assistance, Food assistance, Transportation assistance, Books/equipment assistance, Campus work-study opportunities, Government/institutional schemes.

Final eligibility must be determined by the relevant institution or scheme.

4.3 Well-being Support

Potential signals: Self-reported mood, Stress, Sleep quality, Energy, Academic pressure, Homesickness, Loneliness, Burnout indicators, Voluntary wellbeing assessments.

Potential interventions: AI Support Space, Counsellor appointment, Well-being resources, Support circles, Peer support, Follow-up.

Nivara must not present these signals as medical diagnoses.

5. User Roles

- **Student:** Register/login, Manage consent, View/correct profile information, Complete check-ins, View personal trends, View support recommendations, Understand why recommendations were generated, Explore academic/financial/well-being support, Chat with AI Support Space, Find counsellors, Book counselling sessions, Join temporary support circles, Access resources, Manage notifications, Manage privacy settings.
- **Counsellor:** Manage availability, View appointments, Accept/reject requests, View relevant student support information, Review assessments, View appropriate risk/support alerts, Communicate with students, Add private session notes, Schedule follow-ups, Mark interventions as completed.
- **Administrator:** Monitor campus-level support demand, View aggregated academic/financial/well-being trends, Manage support programs, Manage resources, Manage counsellors, Monitor fairness/bias metrics, Monitor system performance, Configure support rules. *(No unrestricted access to individual private counselling records.)*

6. Student Dashboard Structure

```
Student
├── Today
├── Check-In
├── Academics
├── Insights
├── Support Space [AI]
├── Counsellors
├── Financial Support
├── Resources
├── Temporary Support Circles
├── Multilingual
└── My Profile
```

7. Today Module

The Today page acts as the student's starting point: Check-in status, Upcoming counselling appointment, Pending support actions, Recommended resources, Follow-up reminders, New notifications.

API: `GET /api/student/today`

```
{
  "checkInCompleted": false,
  "upcomingAppointment": null,
  "recommendedActions": [
    "Complete today's check-in"
  ],
  "notifications": 2
}
```

8. Consent Management

Consent is a fundamental part of Nivara. Before collecting optional information, students understand what is collected, why, how it is used, and who can access it.

APIs: `POST /api/consent`, `GET /api/consent`, `PUT /api/consent/:type`

9. Student Profile & Data Correction

Satisfies PS-29 requirement for students to view and correct key information.

[View](#) → [Edit](#) → [Confirm](#)

For institution-controlled information: *Student reports error* → *Correction request submitted* → *Institution verifies & updates* → *Immutable audit log recorded*.

10. Daily Check-In

Lightweight 1–5 ratings: Mood (1-5), Stress (1-5), Sleep (1-5), Energy (1-5), Academic Pressure (1-5).

APIs: `POST /api/checkins` , `GET /api/checkins/history`

11. Academic Support Engine

Identifies academic support needs from signals (stress, missed assignments, attendance trend, exam pressure) → calculates Need Level (MODERATE/HIGH) → recommends advisors, tutoring, workshops.

12. Financial Support Navigator

Minimal data only: Fee difficulty (None to Significant), Expense category (Tuition, Hostel, Food, Books, Transport, Exam), Current support status (Receiving, Not receiving, Applied). **Strictly no bank statements or credit scores.**

13. Financial Support Matching

Matches consented student responses against support program database rules. Phrased as: "*You may want to explore this support option.*"

14. Financial Support Categories

Scholarships, Fee Assistance, Emergency Funds, Hostel Assistance, Food Assistance, Transportation Assistance, Books & Equipment, Campus Work Opportunities, Government Schemes.

APIs: `GET /api/financial-support` , `GET /api/financial-support/:id` , `POST /api/admin/financial-support` , `PUT /api/admin/financial-support/:id`

15. Explainability Engine

Every recommendation answers "*Why am I seeing this?*" with factors, time windows, and reassurance that the recommendation is non-punitive and does not affect grades or disciplinary records.

API: `GET /api/recommendations/:id/explanation`

16. Unified Support Assessment

Combines Academic, Financial, and Well-being signals into a dynamic **Support Profile** (e.g. Academic: Moderate, Financial: High, Well-being: Moderate).

17. Risk/Need Detection Engine

Predefined rules calculating support indicators (e.g. repeated stress + declining sleep → well-being support indicator; fee difficulty + no aid → financial support indicator).

18. No Automated Punitive Decisions

Mandatory System Principle:

Nivara must NEVER automatically reduce grades, cancel scholarships, block registration, flag for disciplinary action, restrict hostel access, or deny services.

All high-impact decisions require human/institutional review.

19. AI Support Space

Layered pipeline: Student → Safety Layer → AI Service → Response Validation → Student. Provides supportive empathetic dialogue, detects safety concerns, and routes to emergency helplines without providing medical diagnosis.

20. Counsellor System & Dashboard

Schedule management, pending requests, assessment overview, and risk/support alerts.

APIs: `GET /api/counsellor/dashboard` , `GET /api/counsellor/schedule` , `GET /api/counsellor/appointments/pending` , `GET /api/counsellor/followups`

21. Appointment System

Lifecycle: `REQUESTED → PENDING → ACCEPTED → COMPLETED → FOLLOW-UP → CLOSED` .

22. Session Notes

Private counsellor notes with strict role & ownership access control.

23. Temporary Support Circles

Time-bounded peer groups for First-Year Homesickness, Exam Stress, Placement Anxiety, Hostel Life, Burnout.

24. AI Moderation

Automated moderation screening for harassment, abuse, spam, and crisis indicators supporting human review.

25. Resources Library

Academic, Financial, and Well-being curated resources (`GET /api/resources`).

26. Admin Dashboard & 27. Early Support Dashboard

Aggregated demand analytics emphasizing campus support demand rather than punitive ranking.

28. Bias & Fairness Dashboard & 29. Monitoring Architecture

Measures Recommendation Rate, Selection Rate, Equal Opportunity Difference, and Disparate Impact across student groups for audit purposes.

30. Student Data Transparency & 31. Data Correction Workflow

Full transparency of stored attributes and an institutional correction request workflow (`POST /api/profile/correction-request`).

32. Database Collections & 33. Key Models

Users, Student Profiles, Counsellor Profiles, Consents, Check-ins, Academic Signals, Financial Profiles, Support Needs, Recommendations, Programs, Appointments, Session Notes, Circles, Notifications, Fairness Metrics, Correction Requests, Audit Logs.

34. API Architecture Summary

Full suite across Auth, Student, Financial, Counsellors, Appointments, AI, Circles, Corrections, and Admin.

35. Backend Folder Structure

```
nivara-backend/  
├── src/  
│   ├── controllers/  
│   ├── models/  
│   ├── services/  
│   ├── middleware/  
│   ├── routes/  
│   ├── config/  
│   └── server.js  
├── .env  
├── package.json  
└── README.md
```

36. Technology Stack

Layer	Technology
Backend	Node.js + Express.js
Database	MongoDB / Mongoose
Authentication	JWT + bcrypt
Real-time	Socket.IO
AI Service	LLM API / Intent Engine

37. Security Requirements & 38. Privacy Principles

Password hashing, JWT security, RBAC, input validation, rate limiting, audit logs, data minimization, consent-first, purpose limitation, and human oversight.

39. MVP for SIH & 40. End-to-End Flow

Identify → Explain → Support → Connect → Follow Up → Improve

41. Nivara's Strongest Differentiators

- 1. Three-Dimensional Support (Academic + Financial + Well-being)
- 2. Explainable Support ("Why am I seeing this?")
- 3. Consent-First Design
- 4. Fairness Monitoring
- 5. Non-Punitive Architecture

42. Final Technical Proposition

"Nivara is an explainable, consent-first student support system that identifies potential academic, financial, and well-being needs from minimal data, connects students to appropriate support, continuously monitors fairness, and prevents those signals from being used for automated punitive decisions."